DOCUMENT 1 OF 5

ELECTRICAL SCHEMATIC AND MASTER NETLIST

Circuit Interconnection Architecture, Full Netlist and Actuator Notes

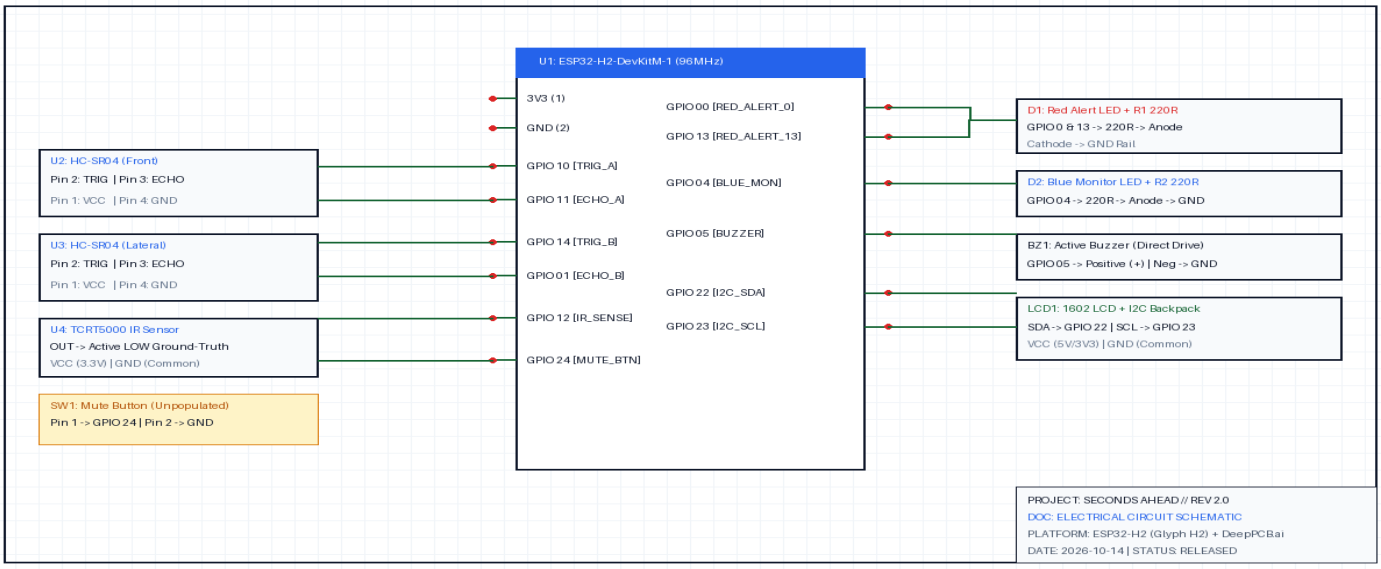


Figure 1.1: *SECONDS AHEAD System Electrical Schematic Block Interconnect Diagram.*

System Interconnection Netlist (Complete Firmware & Hardware Net Table)

Signal / Net Name	GPIO Pin	Connected Component & Pin	Voltage Level	Electrical Function / Mode
TRIG_A	GPIO 10	U2 (Front HC-SR04), Pin 2 [TRIG]	3.3 V Logic	10 µs acoustic transmit trigger pulse (OUTPUT)
ECHO_A	GPIO 11	U2 (Front HC-SR04), Pin 3 [ECHO]	3.3 V Logic*	Acoustic flight return pulse width (INPUT, 25ms timeout)
TRIG_B	GPIO 14	U3 (Lateral HC-SR04), Pin 2 [TRIG]	3.3 V Logic	12ms interleaved lateral ping trigger (OUTPUT)
ECHO_B	GPIO 01	U3 (Lateral HC-SR04), Pin 3 [ECHO]	3.3 V Logic*	Lateral acoustic flight return pulse (INPUT)
IR_CONTACT	GPIO 12	U4 (TCRT5000 IR Sensor), Pin OUT	3.3 V Logic	Active LOW physical contact detector (INPUT_PULLUP)
MUTE_BTN	GPIO 24	SW1 (Mute Push Button), Pin 1	3.3 V Logic	Push-to-GND audio mute toggle (INPUT_PULLUP)
RED_ALERT_0	GPIO 00	D1 (Red 5mm LED), Anode via R1	3.3 V Logic	Primary driver for HIGH_RISK strobe (OUTPUT)
RED_ALERT_13	GPIO 13	D1 (Red 5mm LED), Anode via R1	3.3 V Logic	Redundant parallel drive for Red LED (OUTPUT)
BLUE_MONITOR	GPIO 04	D2 (Blue 5mm LED), Anode via R2	3.3 V Logic	Active HIGH indicator for MONITOR approach state (OUTPUT)
BUZZER_ACT	GPIO 05	BZ1 (Active Buzzer), Positive Terminal	3.3 V Logic	150ms periodic audio beep on HIGH_RISK (OUTPUT)
I2C_SDA	GPIO 22	LCD1 (1602 I2C Backpack), Pin SDA	3.3 V Logic*	I2C Serial Data Bus (400kHz Fast Mode / Standard)
I2C_SCL	GPIO 23	LCD1 (1602 I2C Backpack), Pin SCL	3.3 V Logic*	I2C Serial Clock Line (30ms hardware timeout configured)
+3V3	3V3 Pin	ESP32-H2 VDD, Sensors, LEDs, LCD	3.3 V DC	Main system regulated logic power rail (max 500mA)
GND	GND Pin	Common ground reference bus	0.0 V Ref	System-wide common reference plane

DOCUMENT 1 OF 5 (CONTINUED)

SYSTEM WIRING DIAGRAM AND HARNESS SCHEDULE

Point-to-Point Hardware Wiring Coordinates, Wire Directionality and Power Distribution

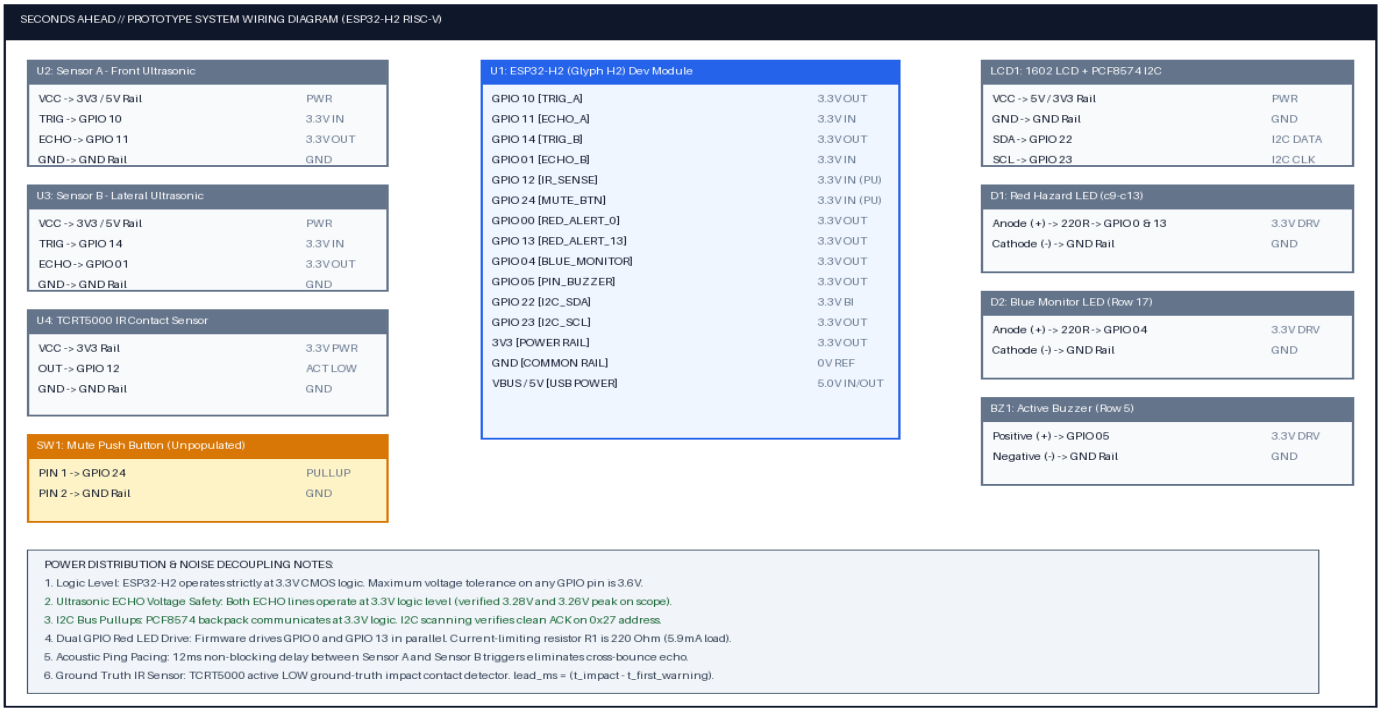


Figure 1.2: *SECONDS AHEAD Prototype Point-to-Point System Wiring Block Diagram.*

Point-to-Point Hardware Wiring Schedule

Origin Node	Wire Label / Signal	Destination Node	Wire Direction	Logic Level	Breadboard Row / Location
ESP32-H2 Pin 10	**TRIG_A**	Front HC-SR04 Pin 2	ESP32 -> Sensor	3.3 V Logic	Yellow wire -> Row 10
ESP32-H2 Pin 11	**ECHO_A**	Front HC-SR04 Pin 3	Sensor -> ESP32	3.3 V Logic*	White wire -> Row 11
ESP32-H2 Pin 14	**TRIG_B**	Lateral HC-SR04 Pin 2	ESP32 -> Sensor	3.3 V Logic	Green wire -> Row 14
ESP32-H2 Pin 01	**ECHO_B**	Lateral HC-SR04 Pin 3	Sensor -> ESP32	3.3 V Logic*	Blue wire -> Row 1
ESP32-H2 Pin 12	**PIN_IR**	TCRT5000 Pin OUT	Sensor -> ESP32	3.3 V Logic	Brown wire -> Row 12 (PU)
ESP32-H2 Pin 0/13	**RED_ALERT**	Red LED Anode via 220R	ESP32 -> LED	3.3 V Logic	Row 9 jumper to c9-c13
ESP32-H2 Pin 04	**BLUE_MONITOR**	Blue LED Anode via 220R	ESP32 -> LED	3.3 V Logic	Row 17 jumper to LED
ESP32-H2 Pin 05	**PIN_BUZZER**	Active Buzzer Positive	ESP32 -> Buzzer	3.3 V Logic	Row 5 jumper to Buzzer
ESP32-H2 Pin 22	**I2C_SDA**	LCD Backpack Pin SDA	Bidirectional	3.3 V Logic*	Yellow wire -> I2C SDA
ESP32-H2 Pin 23	**I2C_SCL**	LCD Backpack Pin SCL	ESP32 -> LCD	3.3 V Logic*	Orange wire -> I2C SCL
ESP32-H2 3V3	**+3V3 Rail**	All Module VCC Pins	Power Supply	3.3 V DC	Red power rail strip
ESP32-H2 GND	**GND Rail**	All Module GND Pins	Common Return	0.0 V Ref	Blue ground rail strip